

Modélisation complète de la base de données

Plateforme nationale de gestion des auto-écoles & examens (Gabon)

Modèle conceptuel + logique orienté B2G, conformité, traçabilité, auditabilité Objectif :
base robuste, scalable, interopérable avec systèmes étatiques

1. ENTITÉS INSTITUTIONNELLES

ministry

- id (UUID)
- name
- code
- contact_email
- contact_phone
- created_at

dgtt

- id (UUID)
- name
- region
- contact
- created_at

cnepc

- id (UUID)
 - name
 - region
 - contact
 - created_at
-

2. AUTO-ÉCOLES

driving_school

- id (UUID)
- legal_name
- commercial_name
- approval_number

- approval_issue_date
- approval_expiry_date
- status (active/suspended/closed)
- address
- city
- gps_lat
- gps_lng
- capacity
- created_at

driving_school_documents

- id
 - driving_school_id (FK)
 - doc_type (agrément, assurance, autorisation, inspection)
 - file_url
 - issue_date
 - expiry_date
-

3. VÉHICULES

training_vehicle

- id
 - driving_school_id (FK)
 - plate_number
 - vehicle_type
 - insurance_status
 - technical_control_status
 - gps_enabled (bool)
 - active
-

4. MONITEURS

instructor

- id
- full_name
- national_id
- professional_id
- certification_level
- driving_school_id (FK)
- status
- created_at

instructor_training

- id
 - instructor_id (FK)
 - training_type
 - institution
 - date
-

5. CANDIDATS

candidate

- id
- full_name
- national_id
- birth_date
- phone
- email
- driving_school_id (FK)
- registration_date
- status

candidate_training

- id
 - candidate_id (FK)
 - theory_hours
 - practical_hours
 - attendance_rate
 - last_update
-

6. EXAMENS

exam_center

- id
- name
- location
- gps_lat
- gps_lng

exam_session

- id
- exam_center_id (FK)
- session_date

- exam_type (theory/practical)
- status

exam_candidate

- id
- exam_session_id (FK)
- candidate_id (FK)
- result (pass/fail)
- score
- attempt_number

examiner

- id
- full_name
- professional_id
- status

exam_assignment

- id
 - exam_session_id (FK)
 - examiner_id (FK)
-

7. CONFORMITÉ & CONTRÔLE

inspection

- id
- driving_school_id (FK)
- institution (DGTT/CNEPC)
- inspection_date
- result
- remarks

sanction

- id
 - driving_school_id (FK)
 - sanction_type
 - reason
 - start_date
 - end_date
-

8. ANALYTICS & AUDIT

performance_metrics

- id
- entity_type (school/instructor/candidate)
- entity_id
- success_rate
- failure_rate
- anomaly_score
- last_update

anomaly_log

- id
 - entity_type
 - entity_id
 - anomaly_type
 - severity
 - detected_at
-

9. SÉCURITÉ & TRAÇABILITÉ

system_user

- id
- name
- role (admin/state/inspector/school)
- institution
- email
- phone
- created_at

access_log

- id
 - user_id (FK)
 - action
 - entity
 - entity_id
 - timestamp
-

10. INTEROPÉRABILITÉ ÉTAT

external_registry_link

- id
 - external_system (DGT/ID_NATIONAL)
 - local_entity
 - local_entity_id
 - external_id
 - sync_status
 - last_sync
-

11. RELATIONS CLÉS

- driving_school 1—N instructor
 - driving_school 1—N candidate
 - driving_school 1—N training_vehicle
 - candidate 1—N exam_candidate
 - exam_session 1—N exam_candidate
 - exam_session N—N examiner
 - driving_school 1—N inspection
 - driving_school 1—N sanction
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12. PRINCIPES ARCHITECTURAUX

- UUID global
 - Soft delete
 - Historisation
 - Journalisation complète
 - Séparation données métier / données audit
 - Prêt API REST / GraphQL
 - Compatible microservices
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13. MODULES APPLICATIFS

- Module Auto-écoles
- Module Moniteurs
- Module Candidats
- Module Examens
- Module Conformité
- Module Inspection
- Module Analytics
- Module Interopérabilité État

14. EXTENSIONS FUTURES COMPATIBLES

- Biométrie (avec base légale)
- Identité numérique
- QR/NFC permis
- Interfaçage police
- Interfaçage assurances

15. CONFORMITÉ STRATÉGIQUE

- Alignement digitalisation titres
- Alignement lutte anti-fraude
- Alignement modernisation transport
- Alignement gouvernance publique

Document prêt pour : - Maquettage UML - Schéma SQL - Architecture API - Dossier projet institutionnel - Prototype MVP